CLASS 8 — MATHS · FORMULA SHEET

Chapter 7: **Proportional Reasoning-1** | Quick revision, one page

Ratio

- Ratio of a to b (same kind, same unit): $a : b = \frac{a}{b}$, $b \neq 0$.
- Simplest form: divide both terms by their HCF. Equivalent ratios: $a : b = ka : kb$ ($k \neq 0$).
- Convert to the same unit before forming a ratio (e.g. 1 m : 40 cm = 100 : 40 = 5 : 2).

Proportion

- $a : b = c : d$ (written $a : b :: c : d$) are in proportion if $ad = bc$ (a, d extremes; b, c means).
- Comparing ratios: convert to fractions with a common denominator, or cross-multiply.

Unitary method

- Find the value of one unit, then multiply: if n items cost C, one costs $\frac{C}{n}$ and m items cost $\frac{C}{n} \times m$.
- Direct variation: the quotient $\frac{y}{x}$ is constant.

Sharing in a ratio

- Share a quantity T in the ratio $p : q$: first part $= \frac{p}{p+q}T$, second $= \frac{q}{p+q}T$.
- Combining ratios: if $a : b$ and $b : c$ are given, make the b terms equal to get $a : b : c$.
- Speed $= \frac{\text{distance}}{\text{time}}$, rate $= \frac{\text{amount}}{\text{one unit}}$ (a ratio with units).

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